Ram Calculus for Hilbert Transforms Along Lipschitz Directions

M169: Rank-One Geometry, Endpoint Outer Scales, and Complete Physical Certificates

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Abstract

The OpenAI manuscript M169 claims a uniform strong L^2 estimate for short Hilbert transforms along arbitrary planar Lipschitz unit fields. Its central difficulty is a scale-independent oscillatory estimate, not the representation of the operator by finite coefficients. We relate that proof structure to the physical-metric, derivative-channel and full-residual methods in the Ram corpus. A concrete refinement uses a a constraint unused in the earlier local certificate: the derivative of a planar unit field has rank at most one. This improves pullback estimates and gives a finite absolute-difference certificate at the outer scale $a\,\mathrm{Lip}(v) = 1$, where the earlier general-field certificate diverged. Explicit band and Sobolev bounds, actual almost-everywhere principal values on a stated domain, inner-cutoff errors, and complete reconstruction certificates are proved. Conditional on the source's uniform short-scale theorem, the outer scale extends to $1/\,\mathrm{Lip}(v)$ on all of L^2, with an explicit additional norm cost. A Fourier-tail Dini domain strictly enlarges the earlier dyadic sufficient domain and supports a restricted maximal bound. A general summable measure-ratio criterion weakens the logarithmic exponent required in the source's restricted-to-strong step. Finite phase metrics and error accumulation are analyzed without replacing exact constant-one estimates by sampled evidence. The independent deductions do not certify the source's entire hierarchy proof, give an unconditional new solution of Stein's conjecture, or establish maximal truncation on every L^2 input.

Contents

1 Pinned Problem and Scope of the Investigation **3**
- 1.1 The exact operator . . . 3
- 1.2 Provenance and mathematical status . . . 3
- 1.3 Relevant corpus and what it actually supplies . . . 3

2 Current Primary Literature and the Source's Position **4**

3 The Source Proof and Its Points of Contact **4**
- 3.1 Dependency structure . . . 4
- 3.2 Why the source's uniform gain is substantial . . . 5

4 Rank-One Geometry Improves the Earlier Pullback Estimate **5**

5 Explicit Difference Bounds and a Finite Endpoint Certificate **6**
- 5.1 A regularity tradeoff down to integrable derivatives . . . 7
- 5.2 A saturated bound on the entire frequency ball . . . 8

6 Actual Full-Frequency Principal Values on a Stated Domain **8**
- 6.1 Sobolev constants and what is still missing . . . 9

7 A Strictly Larger Fourier-Tail Domain **10**
- 7.1 Quantitative positive-regularity continuity . . . 11
- 7.2 A restricted maximal-truncation conclusion . . . 11

8 A Conditional Larger-Scale Consequence of M169 **11**

9 Ram Coefficients Without Changing the Operator **12**
- 9.1 Exact weak coefficients and the symbol . . . 12
- 9.2 A nonlinear straightening is not an exact reduction . . . 13

10 Complete Physical Reconstruction and Tail Accounting **13**

11 A More General Restricted-to-Strong Summation Criterion **14**

12 Finite Phase Metrics and Depth-Safe Error Budgets **14**
- 12.1 Nested averaging without an independence assumption . . . 15
- 12.2 An explicit strict-gain test and its numerical limitation . . . 16

13 Pros, Cons, and Future Research **16**
- 13.1 Advantages established here . . . 16
- 13.2 Costs and unresolved implications . . . 16
- 13.3 Concrete next investigations . . . 16

14 Conclusion **17**

A Source and Verification Register **17**

1 Pinned Problem and Scope of the Investigation

1.1 The exact operator

Let $v : \mathbb{R}^2 \to S^1$ be Lipschitz, $L = \mathrm{Lip}(v)$, and $a > 0$. With Lebesgue measure and no boundary conditions at a finite domain, define, for $0 < h < a$,

$$T_{v,h,a} f(x) = \int_{h<|t|<a} f(x - tv(x)) \frac{\mathrm{d}t}{t}. \tag{1}$$

The field is frozen at the output x. This is not integration along the flow of v. The input space in the conjecture is all of $L^2(\mathbb{R}^2)$; the initial classical domain is $\mathcal{S}(\mathbb{R}^2)$. There is no gauge choice. A unitary dilation is $U_b f(x) = b f(bx)$ and $v_b(x) = v(bx)$. Direct substitution gives

$$T_{v_b,h/b,a/b} U_b = U_b T_{v,h,a}, \qquad \mathrm{Lip}(v_b) = bL.$$

Thus aL is the invariant outer-scale parameter. For $L = 0$ the field is constant and every finite a is permitted.

The source [1], Theorem 1.1 in its introduction, asserts absolute $a_* \in (0, 1/2)$ and $C_* < \infty$ for which

$$\sup_{0<h<a_*/L} \left\| T_{v,h,a_*/L} f \right\|_2 \le C_* \|f\|_2 \quad (L > 0,\ f \in \mathcal{S}). \tag{2}$$

The supremum is outside the norm. It is not the estimate $\|\sup_h |T_{v,h,a} f|\|_2 \lesssim \|f\|_2$. The latter would have additional pointwise consequences. A bounded extension, a strong limit of all inner cutoffs, and an almost-everywhere principal value are also different conclusions. This report keeps them separate at each step.

1.2 Provenance and mathematical status

The source is the September 25, 2026 manuscript, catalogue item 169, family 83, at repository commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The snapshot is fixed, rather than silently following later changes. The local applicability rating is L5 because the earlier Ram report TR-item-355 [12] treats precisely this operator and its missing uniform quantifier. L5 measures applicability, not correctness or novelty.

The source's theorem is treated as a named premise when used. The rank-one estimates, error bounds, summation criterion and implications below are separately proved. They use classical change of variables, Rademacher differentiation, Fourier theory, and the ordinary one-dimensional Hilbert theorem as declared inputs. None of these classical facts is relabeled as a newly discovered Ram theorem.

1.3 Relevant corpus and what it actually supplies

The principal local antecedent is TR-item-355, which proves pullbacks for general bounded Lipschitz fields, band-dependent error bounds, a logarithmically weighted dyadic domain, and physical reconstruction certificates. It explicitly does not establish the uniform L^2 target. TR216 [13] supplies the exact differentiated-atom convention and warns that bounded values of a symbol do not supply all symbol seminorms. The physical-metric passage of RamCalBooks volume 21 [14] retains J^*J and the orthogonal omitted image. Its convergence chapter separates dense-core consistency

from uniform norm control. We use those principles, not self-adjoint resolvent theorems for this variable-direction operator.

The report preserves the useful earlier estimates while exploiting the additional unit-field geometry. It does not assume that a coordinate change, moment representation, learned approximation, or finite matrix automatically resolves a uniform infinite-dimensional estimate.

2 Current Primary Literature and the Source's Position

This is a bounded October 7, 2026 statement-level literature check, not a new full proof audit of each cited paper. Lacey–Li's memoir [2], Conjecture 1.5, states the natural-scale weak $(2,2)$ problem. Its Theorem 1.15 is an annular weak-L^2 result for measurable directions; Theorem 1.18 assumes a stronger popularity-maximal estimate below L^2 for a strong annular conclusion, and additional $C^{1+\eta}$ regularity for its full-transform conclusion. The Lipschitz Kakeya maximal estimate [5], Theorem 1.4, costs $C\delta^{-1/2}$ in weak L^2 at popularity δ. It is an averaging theorem, not the missing strong Hilbert bound.

Bateman–Thiele [3], Theorem 1, prove full-line L^p estimates for $p > 3/2$ for nonvanishing measurable fields depending on one coordinate. Guo [4], Theorem 1.1 and Corollary 1.3, permits suitable Lipschitz foliations with an explicit transverse-angle condition. Those structural hypotheses are not automatic for every two-coordinate Lipschitz field. Guo's annular variation theorem [6], Theorem 1.2, has weak L^2 and strong L^p, $p > 2$, bounds for the specified smooth dyadic variation; it is not a full-spectrum strong-L^2 hard-cutoff maximal theorem. Guo–Thiele's lacunary model [7] and its finite-order extension by Di Plinio–Parissis [8] exploit additional directional structure, including specified dyadic roundings.

Di Plinio–Guo–Thiele–Zorin-Kranich [9], Corollaries 1.3–1.4, control frequency leakage and reduce a local slope transform to uniform strong vertical-band bounds under a small vertical Lipschitz constant. They do not supply that missing band hypothesis. The source uses a compatible reduction while seeking the new oscillatory saving. Its current general theorem remains a source claim in this report: neither a project listing nor a literature search establishes independent validation. The earlier restricted results are not silently promoted to the general theorem, and related multilinear Hilbert transforms are not substituted for this linear frozen-direction problem.

3 The Source Proof and Its Points of Contact

3.1 Dependency structure

The manuscript localizes the unit field, rotates it, and changes to slope form

$$H^h_{u,a} f(x,y) = \int_{h<|t|<a} f(x-t, y-tu(x,y)) \frac{\mathrm{d}t}{t}, \qquad |u| \le 1, \quad \mathrm{Lip}(u) \le C_{\mathrm{Lip}}.$$

The time change modifies both endpoints. The source controls each resulting annulus, including measurable spatially varying endpoints, by positive pullback bounds. A vertical band has frequency $|\eta|$ comparable to w^{-1}, not to w. Its central restricted estimate is

$$|\langle H^h_{u,a} P_w f, g\rangle| \le C(1 + |\log(|F|/|E|)|)^{-3} \sqrt{|F||E|}, \quad |f| \le \mathbf{1}_F, \quad |g| \le \mathbf{1}_E. \tag{3}$$

Uniformity in h, w, u is essential. Amplitude summation gives strong band bounds; a commutator and Littlewood–Paley argument reassemble them. The subsequent oscillatory hierarchy is intended to establish (3), not to assume it in proving its own inputs.

Source component	Required mathematical role	Ram-compatible check or representation
01–02, reduction	Endpoint annuli, local slope pullbacks, restricted-to-strong summation and band reassembly	Preserve the physical measure, both cutoffs, and the full frequency tail.
03–04, tiles and forest	Ordinary size/density bounds; popular directions; disjoint witnesses	Use the actual packet Gram matrix and the admissible rectangle geometry.
05–06, concentration and count	Every-line moments, variation, and one simultaneous exceptional set	A finite sample of lines or one selected list is insufficient for the quantified count.
07, finite phases	Strict coefficient-exponent gain with operator constant exactly one	Separate phase averaging from coefficient normalization; control any nonorthogonal metric.
08–09, hierarchy and prediction	Eligible mass, spatial prediction, normalized path energies	Track derivative and freezing defects with their mass-loss factors.
10, residuals	Square summation on mass drops and variation of one fixed sequence	Keep stopped residuals and avoid choosing a new sequence after observing the test point.
11, assembly	Shifted interpolation, common-grid averaging, depth-independent saving	Sum all tails and losses before removing finite truncations.

3.2 Why the source's uniform gain is substantial

The popular-rectangle estimate carries an inverse-popularity cost. Geometric control alone does not deliver the logarithmic saving. The finite-phase theorem uses a separated list of at most D frequencies and a probability whose Fourier support is narrow enough. For $p \geq 4$ it asserts

$$\left\| \sum_i a_i e^{i\lambda_i t} \right\|_{L^p(\nu)} \leq \left(\sum_i |a_i|^{r_*} \right)^{1/r_*}, \qquad \frac{1}{r_*} = 1 - \frac{1 + c/\log(2D)}{p}.$$

An extra constant greater than one at every hierarchy level cannot be discarded. The source compares a dominant coefficient with a fourth-moment deficit away from that case, then iterates via mass-normalized energies. The intended final gain dominates fixed polynomial losses in $\log \sqrt{|E|/|F|}$. Ram representation makes those losses inspectable, but does not itself create the required gain.

4 Rank-One Geometry Improves the Earlier Pullback Estimate

Write $P_t^v f(x) = f(x - tv(x))$. A generic planar Lipschitz map only gives $\|P_t^v\|_{2\to 2} \leq (1 - |t|L)^{-1}$. For a unit field this loses one half-power unnecessarily.

Lemma 4.1 (Unit-field Jacobian)**.** *At almost every differentiability point of a Lipschitz unit field,*

$$\operatorname{rank} Dv \leq 1, \qquad \det Dv = 0, \qquad |\operatorname{div} v| \leq L.$$

For $|t|L < 1$, $F_t(x) = x - tv(x)$ is a bi-Lipschitz bijection and

$$\det DF_t = 1 - t \operatorname{div} v > 0, \qquad \|P_t^v f\|_2 \leq (1 - |t|L)^{-1/2} \|f\|_2 . \tag{4}$$

For any constant vector e and $0 \leq s \leq 1$, the same conclusions for $F_{t,s}(x) = x - t[e + s(v(x) - e)]$ have sL in place of L.

Proof. Differentiate $|v|^2 = 1$ at a Rademacher point: $v^T Dv = 0$. Every column therefore lies in the one-dimensional space $v^\perp$. Write $Dv = v^\perp b^T$ with $|b| = \|Dv\|_{\mathrm{op}} \le L$. Its trace is $b \cdot v^\perp$, of absolute value at most L. The planar determinant identity gives the displayed Jacobian. The inequality

$$(1 - |t|L)|x - x'| \le |F_t(x) - F_t(x')| \le (1 + |t|L)|x - x'|$$

proves injectivity and inverse Lipschitz control. The equation $x = y + tv(x)$ is a contraction for every y, proving surjectivity. Change of variables now uses the exact one-dimensional determinant loss rather than a product of two worst-case singular values. It yields $\int |f(F_t(x))|^2 \, \mathrm{d}x \le (1-|t|L)^{-1} \|f\|_2^2$. Taking square roots proves (4). The chord field has derivative sDv, so the same argument applies even though the chord field need not have unit length. □

The use of a constant comparison field is important. A chord between two arbitrary variable unit fields can have rank two, so this proof does not assert the same improvement for every field approximation. Bi-Lipschitz inverse images preserve null sets. On every compact t interval with $|t|L < 1$, density of continuous compactly supported functions proves strong continuity of these pullbacks on L^2.

Proposition 4.2 (Critical outer scale is integrable)**.** *For $L > 0$, $0 < a \le L^{-1}$ and every $h > 0$, (1) is a bounded Bochner integral on L^2. The endpoint map at $t = \pm L^{-1}$ need not be invertible and is not used. In particular*

$$\|T_{v,h,a}\|_{2\to 2} \le 2 \int_h^a \frac{\mathrm{d}t}{t\sqrt{1 - tL}} < \infty.$$

Proof. The only new endpoint issue is the integrable singularity $(1 - tL)^{-1/2}$. Strong measurability holds for $|t| < L^{-1}$ by the preceding continuity. Integrate its norm and use completeness. Fubini on bounded spatial sets identifies the Bochner integral with the ordinary pointwise integral for almost every x. The measure-zero endpoints do not affect it. The logarithmic divergence as $h \downarrow 0$ remains; this proposition alone is not a uniform singular-integral theorem. □

5 Explicit Difference Bounds and a Finite Endpoint Certificate

For $0 \le b \le 1$ define

$$J(b) = \frac{2}{1 + \sqrt{1 - b}}, \qquad D(b) = \int_0^1 J(bu) \, \mathrm{d}u. \tag{5}$$

An elementary integration gives

$$D(b) = \frac{4}{b}\left[1 - \sqrt{1 - b} + \log \frac{1 + \sqrt{1 - b}}{2}\right] \quad (b > 0), \qquad D(0) = 1. \tag{6}$$

Thus $1 \le D(b) \le D(1) = 4(1 - \log 2)$, approximately 1.22741. The exact expression, not the decimal, is used in all certificates.

Theorem 5.1 (Absolute comparison with a constant field)**.** *Let v be unit Lipschitz, let e be any constant vector, and put $\delta = \|v - e\|_\infty$. For $|t|L < 1$ and $f \in H^1(\mathbb{R}^2)$,*

$$\|P_t^v f - P_t^e f\|_2 \le \min\left\{|t|\delta J(|t|L) \, \|\nabla f\|_2 \, , [1 + (1 - |t|L)^{-1/2}] \, \|f\|_2\right\}. \tag{7}$$

For $q = aL \le 1$, with the constant-field case interpreted by $q = 0$,

$$\left\| \int_{0<|t|<a} \frac{|P_t^v f - P_t^e f|}{|t|} \, \mathrm{d}t \right\|_2 \le 2a\delta D(q) \, \|\nabla f\|_2 \, . \tag{8}$$

Both the pointwise absolute integral and the absolute Bochner difference are well defined. In particular the symmetric principal value for v exists strongly and almost everywhere on H^1, including $aL = 1$, and

$$\|H_{v,a} f\|_2 \le 2aD(q) \|\nabla f\|_2 . \tag{9}$$

Proof. For smooth f the segment identity is

$$P_t^v f - P_t^e f = -t \int_0^1 (v - e) \cdot \nabla f(F_{t,s}) \, \mathrm{d}s.$$

Lemma 4.1 bounds its norm by $|t|\delta \|\nabla f\|_2 \int_0^1 (1 - s|t|L)^{-1/2} \, \mathrm{d}s$. This integral equals $J(|t|L)$, including its limit at zero. The triangle inequality for the endpoint pullbacks proves the other bound. Approximation in H^1 passes both bounds to general f. Minkowski over the two signs of t, followed by monotone convergence for the nonnegative integral, proves (8). The integral of $J(tL)$ is $aD(q)$, finite even for $q = 1$. Take $e = 0$, so $\delta = 1$ and the symmetric integral of $f(x)/t$ is zero. Absolute integrability gives the pointwise limit; dominated convergence with the L^2 absolute majorant gives the strong limit. This proves (9). No assertion about a noninvertible endpoint map is required. □

5.1 A regularity tradeoff down to integrable derivatives

The determinant calculation has a useful extension beyond the Hilbert space setting. It also clarifies which endpoint is geometric and which is a singular-integral question.

Corollary 5.2 ($W^{1,p}$ principal values at the critical outer scale)**.** *For $1 \le p < \infty$, a unit Lipschitz field and $aL \le 1$, every $f \in W^{1,p}(\mathbb{R}^2)$ has a strong L^p and almost-everywhere symmetric principal value, with*

$$\|H_{v,a} f\|_p \le 2aD_p(aL) \|\nabla f\|_p \le \frac{\pi^2}{3} a \|\nabla f\|_p , \qquad D_p(q) = \int_0^1 \int_0^1 (1 - squ)^{-1/p} \, \mathrm{d}s \, \mathrm{d}u.$$

This is a derivative-domain estimate, not boundedness from L^1 to L^1.

Proof. The exact change of variables gives the L^p pullback constant $(1 - s|t|L)^{-1/p}$ for the chord sv to zero. The segment identity therefore bounds the absolute difference to $f(x)$ by $|t| J_p(|t|L) \|\nabla f\|_p$, where $J_p(b) = \int_0^1 (1 - sb)^{-1/p} \, \mathrm{d}s$. Integrating both time signs gives the asserted $2aD_p(q)$ bound for the pointwise absolute-integral majorant in L^p. Density of smooth functions in $W^{1,p}$ and bounded pullbacks for each interior time justify this argument for the stated domain. The singular endpoint has zero time measure. For $q \le 1$ and $p \ge 1$, Tonelli and the nonnegative geometric series give

$$D_p(q) \le D_1(1) = \int_0^1 \int_0^1 \sum_{n \ge 0} (su)^n \, \mathrm{d}s \, \mathrm{d}u = \sum_{n \ge 0} \frac{1}{(n+1)^2} = \frac{\pi^2}{6}.$$

Thus the absolute majorant is finite even for $p = 1$. Symmetric cancellation of $f(x)/t$ and dominated convergence give the strong and pointwise limits. At $p = 1$, this derivative argument, not a purported integrable all-L^1 pullback bound at the outer endpoint, justifies the finite cutoff on $W^{1,1}$. No claim about $W^{1,\infty}$ density or all-L^1 maximal estimates has been made. □

The independent geometric arguments need only a bounded Lipschitz planar field satisfying $\det Dv = 0$ almost everywhere, rather than $|v| = 1$: rank at most one still gives $|\operatorname{div} v| \le L$. For that larger class the comparison bound retains $\delta = \|v - e\|_\infty$; the bound with comparison zero has an additional factor $\|v\|_\infty$. The unit assumption remains part of the source premise in the conditional all-L^2 theorem below. This distinction prevents an unsupported extension of that source theorem to an arbitrary rank-one field.

5.2 A saturated bound on the entire frequency ball

Use Fourier phase $e^{-ix\cdot\xi}$, so band support in $|\xi| \le R$ implies $\|\nabla f\|_2 \le R\,\|f\|_2$. Define

$$B_q(A) = 2\int_0^1 \min\left\{AuJ(qu),\ 1+(1-qu)^{-1/2}\right\}\frac{\mathrm{d}u}{u}, \qquad 0\le q\le 1, \quad A\ge 0. \tag{10}$$

The value at $q = u = 1$ is interpreted by an improper integral. In the minimum the first term is finite, so no ambiguity affects integration. Put $B_q(0) = 0$.

Corollary 5.3 (Complete band certificate)**.** *For* $\operatorname{supp}\widehat{f} \subset \{|\xi| \le R\}$*, the left side of* (8) *is at most* $B_q(a\delta R)\,\|f\|_2$*. In addition,*

$$B_q(A) \le 2AD(q), \tag{11}$$

$$B_q(A) \le 4\psi(A) + 4\log\frac{2}{1+\sqrt{1-q}}, \tag{12}$$

$$\psi(A) = \begin{cases} A, & 0\le A\le 1,\\ 1+\log A, & A>1. \end{cases}$$

Both bounds hold simultaneously. For $q < 1$ *this improves, or equals, the earlier bound* $2(1-q)^{-1}\Phi(A)$*, where* $\Phi(A) = A$ *for* $A \le 2$ *and* $\Phi(A) = 2 + 2\log(A/2)$ *otherwise.*

Proof. Apply (7) and integrate with $t = au$. Keeping the first term gives (11). For the second, $J(b) \le 2$ and

$$\min\{AuJ(qu), 1+(1-qu)^{-1/2}\} \le \min\{2Au, 2\} + (1-qu)^{-1/2} - 1.$$

The first integral is $4\psi(A)$ after the outer factor two. The second is $4\log[2/(1+\sqrt{1-q})]$, by substituting $z = \sqrt{1-qu}$. This proves (12) at the endpoint as well. For $q < 1$, $J(qu) \le (1-q)^{-1}$ and $1+(1-qu)^{-1/2} \le 2(1-q)^{-1}$. Insert these inequalities inside the minimum to recover the old bound. At $q = 0$, equality gives $B_0(A) = 2\Phi(A)$. These comparisons are inequalities between proved upper bounds, not assertions about the true operator norm. □

6 Actual Full-Frequency Principal Values on a Stated Domain

Fix a constant unit direction e, and let P_0 project onto $|\xi| \le R_0$, P_j onto $2^{j-1}R_0 < |\xi| \le 2^jR_0$ for $j \ge 1$. These are orthogonal Fourier projections. Define

$$b_j = B_q(a\delta 2^jR_0), \quad W_q(f) = \sum_{j\ge 0} b_j\,\|P_jf\|_2, \qquad \mathcal{D}_q = \{f\in L^2 : W_q(f) < \infty\}.$$

For a constant direction, the cutoff multiplier is $-2i\int_h^a \sin(te\cdot\xi)\,\mathrm{d}t/t$ and its limit is $-2i\,\mathrm{Si}(ae\cdot\xi)$. Its modulus is at most six: below one, $|\sin u/u| \le 1$, and integration by parts on $1 \le c \le d$ gives $|\int_c^d \sin u\,\mathrm{d}u/u| \le 2/c$. Splitting an interval at one proves the bound. Plancherel and dominated convergence give strong convergence on all L^2. The classical one-dimensional principal-value theorem and Fubini give almost-everywhere convergence; the removed outer tail is absolutely integrable by Cauchy–Schwarz.

Theorem 6.1 (Difference domain, including the outer endpoint)**.** *For* $q \le 1$ *and* $f \in \mathcal{D}_q$*, the actual absolute difference integral belongs to* L^2 *and has norm at most* $W_q(f)$*. Consequently*

$$H_{v,a}f = H_{e,a}f + \int_{0<|t|<a}(P_t^vf - P_t^ef)\frac{\mathrm{d}t}{t}, \qquad \|H_{v,a}f\|_2 \le 6\,\|f\|_2 + W_q(f).$$

The limit is both strong and almost everywhere, and the same norm bound holds for every inner cutoff. For $Q_N = I - \sum_{j=0}^{N} P_j$,

$$\|H_{v,a} Q_N f\|_2 \le 6 \|Q_N f\|_2 + \sum_{j>N} b_j \|P_j f\|_2 . \tag{13}$$

If $\delta = 0$, *all* b_j *vanish and* $\mathcal{D}_q = L^2$.

Proof. For finite partial sums of f, the preceding band certificate bounds the difference in the Banach space $X = L^2_x(L^1_{\mathrm{d}t/|t|}(0 < |t| < a))$. The weighted summability makes these differences Cauchy in X. To identify the limit with the actual input, restrict to a bounded spatial set and $h \le |t| \le a' < a$. The pullbacks have a uniform L^2 bound on that compact time interval, so L^2 convergence of the inputs gives local product-L^1 convergence of their differences. Convergence in X gives the same local limit. Uniqueness and countable exhaustion in h, a' and space identify the actual difference almost everywhere. This exhaustion deliberately avoids using a squared pullback bound at $t = a = L^{-1}$, which need not be integrable. The X norm supplies the asserted absolute majorant. Dominated convergence proves vanishing inner tails in L^2, and absolute integrability proves the pointwise difference limit. Add the constant-direction theorem. Apply the same result to $Q_N f$ to get (13); no output orthogonality is assumed. □

6.1 Sobolev constants and what is still missing

Take $R_0 = 1$ and $A_0 = a\delta > 0$. Since $\psi(2^j A_0) \le \psi(A_0) + j \log 2$, one can use

$$b_j \le b + dj, \qquad b = 4\psi(A_0) + 4 \log \frac{2}{1 + \sqrt{1-q}}, \quad d = 4 \log 2.$$

For $\sigma > 0$, put $r = 2^{-2\sigma}$ and

$$K_\sigma^2 = b^2 + 2^{2\sigma} \left[\frac{b^2 r}{1-r} + \frac{2bd\, r}{(1-r)^2} + \frac{d^2 r(1+r)}{(1-r)^3} \right]. \tag{14}$$

Orthogonality and Cauchy–Schwarz, with weights $1, 2^{\sigma(j-1)}$, give $W_q(f) \le K_\sigma \|f\|_{H^\sigma}$. The formula follows from the three geometric sums for $1, j, j^2$. This supplies both principal values for every positive Sobolev order, with constants finite at $q = 1$. When $\delta = 0$ use zero instead of this nonsharp expression. Omitted modes have the explicit coefficient

$$K_{\sigma,N}^2 = 2^{2\sigma} r^{N+1} \left[\frac{(b + d(N+1))^2}{1-r} + \frac{2d(b + d(N+1))r}{(1-r)^2} + \frac{d^2 r(1+r)}{(1-r)^3} \right].$$

Thus the sum in (13) is bounded by $K_{\sigma,N} \|Q_N f\|_{H^\sigma}$.

The b_j grow at most logarithmically with frequency, but are not a uniform L^2 bound. Indeed $B_q(A) \ge B_0(A) = 2\Phi(A)$, since both terms in its minimum increase with q. For $\delta > 0$, choose normalized g_j in disjoint dyadic bands and $f = \sum_{j \ge 1} j^{-1} g_j$. Then $f \in L^2$ but $W_q(f) = \infty$. This is a failure of this sufficient certificate, not a counterexample to Stein or to M169. Setting $\sigma = 0$ in (14) is invalid.

7 A Strictly Larger Fourier-Tail Domain

The triangle sum defining W_q can be improved by retaining a physical square norm of the Fourier tail before integrating the scale. For $f \in L^2$ put

$$\omega_2(f,\tau)^2 = (2\pi)^{-2} \int_{\mathbb{R}^2} \min\{\tau^2|\xi|^2, 1\} |\widehat{f}(\xi)|^2 \, \mathrm{d}\xi, \quad E(b) = 1 + (1-b)^{-1/2}, \quad K(b) = \sqrt{J(b)^2 + E(b)^2}.$$

This uses the same physical Plancherel normalization as the report.

Theorem 7.1 (Fourier-tail Dini certificate)**.** *For a unit field, constant unit e, $q = aL \le 1$, and $\delta = \|v - e\|_\infty > 0$, define*

$$Z(f) = 2 \int_0^a K(tL)\omega_2(f, t\delta) \frac{\mathrm{d}t}{t}.$$

If

$$\int_0^{a\delta} \omega_2(f,\tau) \frac{\mathrm{d}\tau}{\tau} < \infty, \tag{15}$$

then $Z(f) < \infty$ and the actual absolute difference majorant has L^2 norm at most $Z(f)$. The principal-value and reconstruction conclusions previously using W_q remain true with Z. Set $Z = 0$ separately for $\delta = 0$.

Proof. For a fixed interior time put $\tau = |t|\delta$ and split $f = g + r$ by the orthogonal Fourier projections onto $|\xi| \le \tau^{-1}$ and its complement. Apply the derivative branch of (7) to g and the endpoint-pullback branch to r. Cauchy–Schwarz in these two terms gives

$$\|P_t^v f - P_t^e f\|_2 \le J(|t|L)\tau \, \|\nabla g\|_2 + E(|t|L) \, \|r\|_2 \le K(|t|L)\omega_2(f,\tau).$$

The fixed-time pullbacks already act on every L^2 input, so Minkowski and monotone convergence apply to their actual difference, not to a formal interchange of two infinite expansions. They give the asserted majorant. Near zero $K(tL)$ is bounded, so (15) suffices. Away from zero $\omega_2 \le \|f\|_2$ and $K(tL) \le \sqrt{8}(1 - tL)^{-1/2}$, integrable even at $q = 1$. Strong and pointwise convergence follow as in Theorem 6.1. □

Proposition 7.2 (Strict enlargement and logarithmic regularity)**.** *For $\delta > 0$, $\mathcal{D}_q$ is contained in the domain (15), and the containment is strict. A sufficient condition for (15) is*

$$\sum_{j \ge 0} (1+j)^{2\sigma} \|P_j f\|_2^2 < \infty \quad \text{for some } \sigma > 1. \tag{16}$$

This is still a condition stronger than merely $f \in L^2$.

Proof. The triangle inequality in the weighted Fourier space bounds $\omega_2(f,\tau)$ by $\sum_j \min\{\tau 2^j R_0, 1\} \|P_j f\|_2$. Integration and Tonelli give an upper bound $\sum_j \psi(a\delta 2^j R_0) \|P_j f\|_2$ for the Dini integral. Since $\psi(A) \le \Phi(A) \le B_q(A)/2$, it is at most $W_q(f)/2$.

To prove (16), it suffices, by monotonicity of ω_2 and a finite upper-scale interval, to sum $\omega_k = \omega_2(f, 2^{-k}/R_0)$ for $k \ge 0$. Orthogonality gives

$$\omega_k^2 \le \sum_{j \le k} 4^{j-k} \|P_j f\|_2^2 + \sum_{j > k} \|P_j f\|_2^2 .$$

Put $\gamma = 2\sigma - 1 > 1$. Weighted Cauchy–Schwarz bounds $(\sum_k \omega_k)^2$ by $[\sum_k (1+k)^{-\gamma}][\sum_k (1+k)^{\gamma} \omega_k^2]$. Exchanging nonnegative sums bounds the second bracket by $C_\gamma \sum_j (1+j)^{\gamma+1} \|P_j f\|_2^2$: the upper tail

uses $\sum_{k<j}(1+k)^\gamma \le C_\gamma(1+j)^{\gamma+1}$, and the lower part uses geometric decay in $k-j$. This proves the sufficient condition.

For strictness choose unit, orthogonal band inputs g_j and $f = \sum_{j\ge1} j^{-7/4} g_j$. Its weighted dyadic sum diverges because b_j grows like j. On the other hand the displayed modulus bound gives $\omega_k^2 = O(k^{-5/2})$, so $\sum_k \omega_k < \infty$. For contrast, coefficients j^{-1} give an L^2 input with $\omega_k^2 \ge \sum_{j>k+1} j^{-2}$, since the high bands are fully above the cutoff. Its Dini sum diverges. Both examples describe certificate domains, not nonexistence of the transform on their complements. □

7.1 Quantitative positive-regularity continuity

For $0 < s < 1$, interpolation between the L^2 and H^1 bounds for $P_t^v - P_t^e$ gives

$$\|(P_t^v - P_t^e)f\|_2 \le 2(|t|\delta)^s(1-|t|L)^{-(1-s)/2}\,\|f\|_{H^s}\,.$$

Indeed the two endpoint constants can be bounded by $2(1-|t|L)^{-1/2}$ and $2|t|\delta$, and the standard Fourier Hilbert scale interpolates exactly. Integrating the positive norm bound yields

$$\|A_f\|_2 \le 4(a\delta)^s \mathrm{B}\left(s, \frac{1+s}{2}\right)\|f\|_{H^s}\,, \qquad aL \le 1, \tag{17}$$

where A_f is the absolute difference integral and B is the Euler beta integral. This gives a quantitative error tending to zero with δ at fixed positive regularity, including the outer endpoint. At $s = 1$ use the sharper derivative estimate; $s = 0$ is not permitted.

7.2 A restricted maximal-truncation conclusion

The classical one-dimensional maximal Hilbert theorem [10], rotation and Fubini give $\|\sup_{0<h<a}|T_{e,h,a}f|\|_2 \le C_{\max}\|f\|_2$, uniformly in constant e, a. Its kernel normalization is $\mathrm{d}t/t$; $C_{\max}$ is not the cutoff multiplier constant six. On either of the certified domains,

$$\sup_{0<h<a}|T_{v,h,a}f| \le \sup_{0<h<a}|T_{e,h,a}f| + A_f.$$

Consequently the maximal norm is at most $C_{\max}\|f\|_2 + W_q(f)$ or $C_{\max}\|f\|_2 + Z(f)$, respectively. The suprema can first be taken over rational cutoffs; absolute continuity in a positive cutoff away from zero identifies the full supremum almost everywhere. This is a genuine restricted-domain maximal assertion, not a general-L^2 consequence of the source's outside-the-norm supremum.

8 A Conditional Larger-Scale Consequence of M169

Theorem 8.1 (Endpoint outer-length extension)**.** *Assume the source estimate (2). For every unit Lipschitz field with $L > 0$, all $0 < a \le L^{-1}$ and $0 < h < a$ satisfy*

$$\|T_{v,h,a}f\|_2 \le \left[C_* + 4\,\mathrm{artanh}\,\sqrt{1-a_*}\right]\|f\|_2\,, \qquad f \in L^2. \tag{18}$$

The inner cutoffs converge strongly on all L^2 to the unique bounded extension of the principal value on H^1. No maximal-truncation or almost-everywhere conclusion for every L^2 input is asserted here.

Proof. Put $a_0 = a_*/L$. When $a \ge a_0$, split off the annulus $a_0 < |t| < a$. Lemma 4.1 gives its norm at most

$$2\int_{a_0}^{a}\frac{\mathrm{d}t}{t\sqrt{1-tL}} = 4\left[\mathrm{artanh}\,\sqrt{1-a_*} - \mathrm{artanh}\,\sqrt{1-aL}\right]. \tag{19}$$

The final term is zero when $aL = 1$. For $h < a_0$ use the source bound plus this annulus. For $h \geq a_0$ the annulus alone suffices. For $a < a_0$, the source cutoff at h minus its cutoff at a gives an a priori $2C_*$ bound, but the desired displayed constant follows more directly by scaling: the source theorem applied to the same field with the larger allowed Lipschitz bound a_*/a gives C_* at outer length a. Equivalently normalize by that bound; the actual seminorm is then at most one. The theorem is uniform over this class. All finite cutoffs are bounded on L^2 by Proposition 4.2; density extends their Schwartz bounds. For $f \in L^2$ approximate by $g \in H^1$. Uniformly in h, k,

$$\|T_{v,h,a}f - T_{v,k,a}f\|_2 \leq 2M \|f - g\|_2 + \|T_{v,h,a}g - T_{v,k,a}g\|_2 ,$$

where M is the constant in (18). Choose g first, then use Theorem 5.1 to let $h, k \downarrow 0$. Completeness and density prove strong convergence and uniqueness. □

This is an extension of a named source premise, not an independent proof of that premise. It shows a concrete advantage of retaining the exact unit-field Jacobian: the generic planar estimate has a nonintegrable outer loss at $aL = 1$. We do not claim that one is the optimal possible outer-scale constant for every field, or that no larger-scale theorem can hold under additional structure.

9 Ram Coefficients Without Changing the Operator

9.1 Exact weak coefficients and the symbol

For $|t|L < 1$, let $G_t = F_t^{-1}$. The physical adjoint is exactly

$$(P_t^v)^* \phi(y) = \frac{\phi(G_t(y))}{1 - t \operatorname{div} v(G_t(y))}. \tag{20}$$

Change of variables proves this for L^2 tests, with the usual sesquilinear inner product. For any fixed $h > 0$, integration gives

$$\langle T_{v,h,a}f, \phi_\lambda \rangle = \left\langle f, \int_{h<|t|<a} (P_t^v)^* \phi_\lambda \frac{\mathrm{d}t}{t} \right\rangle .$$

Compactly supported smooth Ram atoms may be used here. This moves evaluation onto an analytically represented test atom but retains the inverse map and Jacobian. Dropping those factors changes the adjoint. For a nonconstant field one cannot simply assert $H_v^* = -H_v$.

On Schwartz inputs Fourier inversion gives the exact left symbol

$$a_v(x, \xi) = -2i \operatorname{Si}(a\xi \cdot v(x)).$$

For a differentiability direction,

$$\partial_{x_k} a_v = -2ia(\xi \cdot \partial_{x_k} v) \operatorname{sinc}(a\xi \cdot v), \qquad \operatorname{sinc}(0) = 1.$$

Although $|a_v| \leq 6$, near $\xi \cdot v = 0$ the derivative can grow like $|\xi|$. Merely Lipschitz fields do not have every classical symbol derivative. TR216's smooth, properly supported $S^0_{1,0}$ mapping theorem therefore cannot be invoked from a bounded value channel alone. The exact coefficient and the all-frequency norm estimate are separate objects. This is the relevant derivative-channel lesson of Ram calculus.

9.2 A nonlinear straightening is not an exact reduction

The unit field $v(x,y) = (1,x)/\sqrt{1+x^2}$ has $\mathrm{Lip}(v) \le 1$. The frozen line at the origin is $(-t,0)$, whereas its backward flow is $(-t, t^2/2)+O(t^3)$. This follows from $v(0) = (1,0)$ and $Dv(0)v(0) = (0,1)$. The quantity $y - x^2/2$ is constant along the flow but equals $-t^2/2$ on the frozen line. A flow chart therefore needs a trajectory and kernel correction before it can be used in (1). Small local geometric errors still need summable operator bounds across the singular scales.

10 Complete Physical Reconstruction and Tail Accounting

Let $J_0 : \mathbb{C}^N \to H^1$ be injective synthesis, with $G = J_0^* J_0 > 0$ and $G_\nabla = (\nabla J_0)^*(\nabla J_0) \succeq 0$. These are physical matrices: $\|J_0 c\|_2^2 = c^* G c$ and $\|\nabla J_0 c\|_2^2 = c^* G_\nabla c$. They include cross terms.

Proposition 10.1 (Constant-core reconstruction)**.** *Let $f = J_0 c + r$ and suppose an actual reconstructed output y satisfies $\|y - H_{e,a} J_0 c\|_2 \le \zeta$, with e a constant unit direction. For $r \in \mathcal{D}_q$,*

$$\|H_{v,a} f - y\|_2 \le 2a\delta D(q)\sqrt{c^* G_\nabla c} + 6\|r\|_2 + W_q(r) + \zeta. \tag{21}$$

If $\operatorname{ran} J_0$ is contained in $|\xi| \le R$, the first term can be replaced by $B_q(a\delta R)\sqrt{c^* G c}$. If instead $r \in H^1$, replace the middle two terms by $2aD(q)\|\nabla r\|_2$. Under Theorem 8.1's source premise they can be replaced by $M\|r\|_2$ for every $r \in L^2$.

Proof. Use the exact decomposition $H_v f - y = (H_v - H_e)J_0 c + H_v r + (H_e J_0 c - y)$ and the proved bounds. All three terms are defined on the indicated domains. No residual component or input tail is omitted. □

For $0 < h < a$, the actual inner-cutoff error on H^1 satisfies

$$\|H_{v,a} f - T_{v,h,a} f\|_2 \le 2hD(hL)\|\nabla f\|_2 . \tag{22}$$

This is Theorem 5.1 on the shorter interval with comparison zero. The total ζ must also account for quadrature, atom synthesis, metric errors and omitted output. A small fitted coefficient error is not an automatic bound on ζ.

Proposition 10.2 (Omitted-output and metric certificate)**.** *For an orthogonal output projection Q and $T = H_{v,a}$ on the finite synthesis domain, write*

$$K = (TJ_0)^*(TJ_0) = K_Q + K_\perp, \quad K_Q = (QTJ_0)^*(QTJ_0), \quad K_\perp = ((I-Q)TJ_0)^*((I-Q)TJ_0).$$

If $\widehat{G} > 0$, $G \succeq (1-\gamma)\widehat{G}$ with $\gamma < 1$, $K_Q \preceq \widehat{K} + E \preceq \kappa^2 \widehat{G}$, and $K_\perp \preceq \tau^2 \widehat{G}$, then

$$\|TJ_0 c\|_2 \le \sqrt{\frac{\kappa^2 + \tau^2}{1-\gamma}}\,\|J_0 c\|_2 .$$

Proof. The two outputs are orthogonal, giving the identity by Pythagoras and polarization. Add the quadratic-form inequalities and divide by the lower input metric. This proves a bound for every coefficient vector, not just tested samples. □

For a simple failure control take $J_0 c = (c,0)$, $T(x,y) = (0, Mx)$, and $Q(x,y) = (x,0)$. Then $QTJ_0 = 0$ while $\|TJ_0 c\| = |M|\,|c|$. Also $G_t = \mathrm{diag}(1, t^2)$ and $\widehat{G}_t = \mathrm{diag}(1, t^2 + t)$ are entrywise close as $t \downarrow 0$ but have squared norm ratio $1 + 1/t$ on $(0,1)$. Neither example disproves the source; both disprove unsafe finite-to-full or entrywise-to-relative certificate rules.

11 A More General Restricted-to-Strong Summation Criterion

The logarithmic exponent three in (3) is more than the final amplitude summation itself requires. This does not say that the source's hierarchy can be weakened at every earlier use of a power.

Theorem 11.1 (Summable measure-ratio envelope)**.** *Suppose a bilinear form satisfies*

$$|\mathcal{B}(f,g)| \le C_0\omega(|\log(|F|/|E|)|)\sqrt{|F||E|} \quad (|f| \le \mathbf{1}_F,\ |g| \le \mathbf{1}_E)$$

on finite positive-measure sets, where $\omega \ge 0$. Define

$$k_n = \sup_{u\in[(n-1)\log 2,(n+1)\log 2]} \omega(|u|).$$

If $\sum_{n\in\mathbb{Z}} k_n < \infty$, then on finite amplitude sums

$$|\mathcal{B}(f,g)| \le 32C_0 \|k\|_{\ell^1} \|f\|_2 \|g\|_2 . \tag{23}$$

Completion gives the corresponding L^2 form. In particular $\omega(u) = (1+u)^{-\alpha}$ works for every $\alpha > 1$.

Proof. Write $f = \sum_i 2^i f_i$ on disjoint sets $F_i = \{2^{i-1} < |f| \le 2^i\}$; set $a_i = 2^i\sqrt{|F_i|}$. Group indices by $2^m \le |F_i| < 2^{m+1}$ and let $A_m^2 = \sum_{i \text{ in class } m} a_i^2$. A geometric sum from the largest amplitude index shows $\sum_{i \text{ in class } m} a_i \le 2\sqrt{2}A_m$. Moreover $\|A\|_2 \le 2\|f\|_2$. The same construction gives B_n for g. The logarithmic measure ratio lies within the interval defining k_{m-n}. Hence the bilinear sum is at most $8C_0\sum_{m,n} k_{m-n}A_mB_n$. Young's inequality and the two factors of two prove (23). Finite decompositions justify every sum; then use completion or an already continuous finite cutoff. □

For computed, non-translation-invariant nonnegative envelope errors E_{mn}, a weighted Schur certificate also works: if $w_m > 0$ and

$$\sup_m w_m^{-1}\sum_n E_{mn}w_n \le A, \qquad \sup_n w_n^{-1}\sum_m E_{mn}w_m \le B,$$

then $\|E\|_{\ell^2\to\ell^2} \le \sqrt{AB}$. Indeed apply Cauchy–Schwarz to each row using weights $E_{mn}w_n$, then use the column inequality. This includes all measure classes when the displayed sums are certified with tails. A finite principal matrix is insufficient. The scalar positive matrix $E_{mn} = (1+|m-n|)^{-1}$ on $1 \le m,n \le N$ has Rayleigh quotient of order $\log N$ on the constant vector. Thus at exponent one this particular summation method has a real obstruction; additional cancellation could still improve a particular bilinear form.

12 Finite Phase Metrics and Depth-Safe Error Budgets

For a finite list λ_i, write $J_\lambda c(t) = \sum_i c_i e^{i\lambda_i t}$ in $L^2(\nu)$. Its Gram matrix is

$$G_{ij} = \int e^{i(\lambda_j-\lambda_i)t}\,\mathrm{d}\nu(t) = \widehat{\nu}(\lambda_i-\lambda_j).$$

If frequencies are separated by Δ and $\operatorname{supp}\widehat{\nu} \subset (-\Delta/4,\Delta/4)$, then $G = I$ exactly. For a different atom family one must use its physical G; diagonal phase motion is not automatically unitary in an arbitrary metric. For a diagonal phase matrix $D(t)$, the precise condition is $D(t)^*GD(t) = G$. A unitary change to an orthonormal physical basis preserves norms but need not preserve diagonal phase

structure. Both requirements must be checked together before invoking the source's Hilbert-valued constant-one theorem.

The fourth moment is also an exact finite expression:

$$\int |J_\lambda c|^4 \,\mathrm{d}\nu = \sum_{i,j,k,l} c_i c_j \overline{c_k c_l} \widehat{\nu}(\lambda_k + \lambda_l - \lambda_i - \lambda_j).$$

Only approximate additive relations within the Fourier support survive. This permits exact rational or interval checks for a specified finite configuration and measure. It does not verify the universal coefficient inequality by sampling coefficients. The source's argument ranges over all complex vectors, every separated configuration and every finite depth.

If an approximation introduces a factor $1 + \eta_\ell$ at level ℓ, its accumulated cost is at most

$$\prod_{\ell=1}^{N} (1 + \eta_\ell) \le \exp\left(\sum_{\ell=1}^{N} \eta_\ell\right) \quad (\eta_\ell \ge 0).$$

A depth-independent sum gives a controlled factor. A fixed $\eta > 0$ does not: $(1+\eta)^N$ diverges. To preserve a quantified strict phase gain e^{-g_N}, a sufficient explicit budget is $\sum_{\ell \le N} \eta_\ell \le \theta g_N$, $0 \le \theta < 1$, leaving at least $e^{-(1-\theta)g_N}$. This is an elementary but useful acceptance rule for Ram atom, metric and quadrature errors. It supplies no unproved value of g_N and no automatic uniform tail control.

12.1 Nested averaging without an independence assumption

The following elementary formulation makes a key source mechanism available to exact positive-coefficient implementations.

Lemma 12.1 (Nested positive factors)**.** *Let (X, μ) be a finite measure space, and let finite measurable partitions $\mathcal{P}_0, \dots, \mathcal{P}_N$ successively coarsen. Suppose $\Gamma_j \ge 0$ is constant on cells of $\mathcal{P}_{j-1}$ and its mean on every positive-measure cell of $\mathcal{P}_j$ is at most $1 + \epsilon_j$, where $\epsilon_j \ge 0$. Then*

$$\int_X \prod_{j=1}^{N} \Gamma_j \,\mathrm{d}\mu \le \mu(X) \prod_{j=1}^{N} (1 + \epsilon_j).$$

Proof. Every factor with index greater than one is constant on each cell of $\mathcal{P}_1$. Sum the cell integrals of Γ_1 times those nonnegative constants, using the stated mean bound. Repeat with $\mathcal{P}_2$, and continue. Null cells contribute zero. This is a conditional averaging identity and uses no stochastic independence. For countable partitions the same proof follows by Tonelli and nonnegative truncation whenever the cell means are defined. □

The source has $\epsilon_j = 0$ for its normalized prediction factors. Consequently this step has exactly constant one even for arbitrarily many levels. A numerical replacement needs certified cell means, not pointwise samples. If its errors are nonzero, their sum must satisfy the depth budget above.

A related scalar budget is, for $m_0 \ge \cdots \ge m_N > 0$,

$$\sum_{j<N} \frac{m_j - m_{j+1}}{m_j} \le \sum_{j<N} \log \frac{m_j}{m_{j+1}} = \log \frac{m_0}{m_N}.$$

Indeed $1 - r \le -\log r$ for $0 < r \le 1$. Thus errors proportional to relative mass loss can cost a power of a mass ratio rather than a factor per level. A possibly tiny terminal mass must be handled separately, as in the source's stopping rule; this inequality alone does not make the ratio uniformly bounded.

12.2 An explicit strict-gain test and its numerical limitation

The source's final exponent decrease has the form

$$\delta_L = \frac{\theta - (1+\theta)/K}{p}, \qquad \theta = \frac{c_{\mathrm{ph}}}{\log(2D)}, \quad c_{\mathrm{ph}} = 2^{-60}.$$

Here $p > 0$, $D \geq 1$, and K is the shifted hierarchy parameter. If $K \geq 2 + 2/\theta$, then $(1+\theta)/K \leq \theta/2$ and $\delta_L \geq \theta/(2p) > 0$. The source's choices $p = \sqrt{L}$, $D = L^{C_D}$ and K of order $(\log L)^2$ meet this condition eventually for fixed constants. Together with $\log N \geq L/2$, the saving is of size $\exp[-c\sqrt{L}/\log L]$, dominating fixed powers of L. This verifies the asymptotic exponent comparison, not every preceding hierarchy estimate or a practical onset scale.

The constant c_{ph} is extremely small. Forming $2 + \theta$ in ordinary binary64 arithmetic can round it to two and erase the strict gap. A certificate should store θ separately and compare $K\theta$ with $2(1+\theta)$ by directed interval or exact symbolic arithmetic. The source's theorem is asymptotic; a loss of the gap in a naive floating-point computation is not a mathematical contradiction.

13 Pros, Cons, and Future Research

13.1 Advantages established here

The exact unit constraint gives sharper constants and permits a larger outer-scale certificate. Physical synthesis retains nonorthogonal cross terms and omitted output. The weak coefficient formula keeps the inverse-map Jacobian rather than silently replacing frozen lines by flow trajectories. The summable measure-ratio criterion exposes which loss the amplitude summation can actually tolerate. These are specific transparency and controllability improvements, not evidence of a faster implementation that has not been built. The Fourier-tail Dini domain strictly includes the earlier dyadic certificate domain; its logarithmic regularity condition and restricted maximal bound are further proved benefits. The $W^{1,p}$ extension reaches integrable derivatives at $p = 1$ without asserting all-L^1 boundedness.

13.2 Costs and unresolved implications

The source's many-level oscillatory geometry remains essential for its unconditional all-L^2 claim. The independent Ram comparison domain is smaller than all L^2 for a nonconstant comparison. Gram matrices can be dense or ill-conditioned, and physical tails can be expensive to certify. Pointwise bounded symbols, one frequency band, and finite operator sections do not replace scale-uniform estimates. No measured speedup, optimal constant, complete formal validation or new universal maximal theorem is claimed.

13.3 Concrete next investigations

1. Seek a depth-independent full-output certificate for the source's normalized residual sequence, retaining its rule that the sequence is chosen before the spatial point. A successful finite calculation must include an analytic tail theorem.
2. Exploit the exact rank-one Jacobian in localized slope/unit-field comparisons and compute whether it materially reduces certified constants in an implementation. The exponent of the oscillatory gain must not be replaced by a pullback bound.

3. Test orthonormal physical atom choices that preserve diagonal phase motion. Nonorthogonal coordinates may improve approximation while making the constant-one inequality harder to retain.
4. Extend the proved restricted maximal estimate to variation norms on the difference domain, then identify the additional argument needed on all L^2. Strong convergence alone is not such an argument.
5. For larger outer lengths or weaker field regularity, first establish global pullback geometry and the singular-scale uniform estimate. The critical endpoint proved here is not a license to continue through a fold or to transfer results from arbitrary flow operators.

14 Conclusion

M169 is directly relevant to the Ram operator program because it treats the same uniform singular-integral target isolated by TR-item-355. The present investigation provides exact reformulations and concrete extensions of the earlier certificates. In particular, unit-field rank-one geometry gives a finite endpoint outer-scale error budget, actual principal values on a full-frequency regularity domain, and a conditional extension of the source's bound to outer length L^{-1}. The Dini domain and its restricted maximal estimate improve the earlier sufficient domain, while the nested-factor and strict-gain formulas expose the precision and error budgets of the source proof. The source's all-scale oscillatory theorem is kept separate from these proved consequences. The productive role of Ram calculus is to make representation, physical norm, error and limiting quantifiers explicit, while preserving rather than bypassing the difficult analytic estimate.

A Source and Verification Register

The local source directory is

`_sources/openai-math/preprints/A-uniform-Hilbert-transform-estimate-for-Lipschitz-directions-September-25-2026/build`.

An independent source review read all eleven mathematical sections (7,450 source lines) and recorded its dependency, parameter, residual, and limiting checks in `_audit/M169-source-audit.txt`. The authoring review read the introduction, selected reduction and phase passages, and that detailed audit. The source review found no demonstrated actionable error or counterexample within its scope. That is not a formal verification, independent community validation, or a guarantee that no subtle gap remains. The general source theorem therefore retains its explicitly conditional role in this report.

The local corpus and extension checks are recorded in `_audit/M169-literature-local.txt`. The mathematical deductions used here have their own proofs, including their domains and limiting arguments. The supplementary program `_audit/check_M169.py` records 83 diagnostic checks: exact rational rank-one examples, floating-point integral regressions, and adverse metric and depth-accumulation controls. The floating-point regressions are not interval proofs and are not premises of the universal theorems. External literature is used in the limited statement-level scope stated in the text; no exhaustive census or historical-priority claim is made. All source and private corpus originals remain unchanged.

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